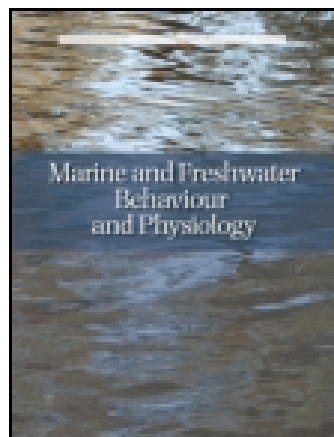




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Within-shelter behaviour of the spotted spiny lobster, *Panulirus guttatus* (latreille), in simulated communal dens: an exploratory study

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WITHIN-SHELTER BEHAVIOUR OF THE SPOTTED SPINY LOBSTER, *PANULIRUS GUTTATUS* (LATREILLE), IN SIMULATED COMMUNAL DENS: AN EXPLORATORY STUDY

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Panulirus guttatus is a sedentary spiny lobster that dwells in the coral reef habitat. We conducted an exploratory study to describe the within-shelter behaviour of groups of 4–6 individuals of *P. guttatus* in simulated communal dens deployed in tanks. Lobsters displayed a diel within-shelter activity rhythm, with a peak shortly before dawn due to an increase in agonistic interactions among sheltered individuals. In a single-chamber cave, aggressions resulted in the constant rearrangement of individuals in the den, related to their preference to cling to the walls. In a triple-chamber cave, the largest male often defended one chamber, but the remaining individuals tended to aggregate in one of the other two chambers during the day. These trends suggest that *P. guttatus* is a gregarious lobster, but that this gregariousness is influenced by the amount of available shelters and by temporary, size-related social hierarchies. This hypothesis should be tested with fully replicated experiments to help understand the patterns of shelter utilization of *P. guttatus* in the coral reef habitat.

Keywords: *Panulirus guttatus*; Behaviour; Gregariousness; Shelter; Den; Diel rhythm; Activity cycle; Agonistic interactions

INTRODUCTION

Spiny lobsters (Palinuridae) occur in high numbers in all tropical and subtropical seas of the world and constitute important fishing resources. Also, palinurids are usually the largest decapods in their habitats and often the largest predator, which makes them ecologically consequential in many marine communities (Phillips, 2000).

Studying the behaviour of marine animals helps to understand how these organisms are adapted to living in complex and changing environments. In particular, information about variation in activity contributes to an understanding of habitat requirements,

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patterns of resource utilisation, and the potential for intraspecific interactions (Smith *et al.*, 1998). The behaviour and distribution of spiny lobsters may influence their catchability by the fishery, and some palinurid species that exhibit gregarious behaviour have been held successfully in communal rearing systems (Phillips, 2000). Hence, the number of behavioural studies on spiny lobsters has steadily increased in recent times, although most have been conducted on a few species of high importance to fisheries. Behavioural issues addressed in such studies include locomotory and feeding activity (e.g., Fielder, 1965; Kanciruk and Herrnkind, 1973; Morgan, 1978; Lipcius and Herrnkind, 1982); predation risk and social interactions (e.g., Berrill, 1975; Cobb, 1981; Lozano-Álvarez and Spanier, 1997); reproductive behaviour (e.g., Berry, 1970; McKoy, 1979; Lipcius and Herrnkind, 1985); den selection and den occupancy (e.g., Lozano-Álvarez and Briones-Fourzán, 2001); homing and movement patterns (e.g., Herrnkind, 1980; Jernakoff, 1987; Lozano-Álvarez *et al.*, 2002), and ontogenic changes in social behaviour (e.g., Childress and Herrnkind, 1996).

Panulirus guttatus, the spotted spiny lobster of the Caribbean, is a small-sized palinurid and a minor fishing resource that remains poorly known, unlike its sympatric species, *P. argus* (the common Caribbean spiny lobster), which is heavily exploited and has been widely studied. The maximum size reported for *P. guttatus* in the Mexican Caribbean is 88 mm carapace length (CL) (Negrete-Soto *et al.*, 2002) and that for *P. argus* is 204 mm CL (Lozano-Álvarez *et al.*, 2003). Individuals of *P. guttatus* appear to live solely in the coral reef habitat during their whole benthic life (Briones-Fourzán and McWilliam, 1997; Sharp *et al.*, 1997), by contrast to *P. argus*, which has several ontogenic habitat shifts. The juvenile phase of *P. guttatus* has seldom been observed, but the adults (females > 38 mm CL, and males > 48 mm CL, Sharp *et al.*, 1997) roam at night in the reef habitat, where they hide in caves and crevices during the day (Caillouet *et al.*, 1971; Negrete-Soto *et al.*, 2002), sometimes sharing dens with *P. argus* (Sharp *et al.*, 1997; Lozano-Álvarez and Briones-Fourzán, 2001). Individuals of *P. argus* are highly mobile and undergo periodic migrations that may span hundreds of kilometers, but those of *P. guttatus* are sedentary (Herrnkind, 1983), with a home range of about 100 m in radius (Lozano-Álvarez *et al.*, 2002).

Many palinurids shelter communally. Such aggregations appear to arise from intrinsic gregariousness (i.e., aggregations that provide advantages to the group of individuals) rather than from a patchy distribution of dens (McCoy and Leachman, 1982; Zimmer-Faust and Spanier, 1987). Moreover, aggregation is mediated by intraspecific chemical attraction in *P. interruptus*, *P. argus* and *Jasus edwardsii* (Zimmer-Faust *et al.*, 1985; Ratchford and Eggleston, 1998; Butler *et al.*, 1999). This has not been determined in *P. guttatus*.

Palinurids, like many decapods, exhibit rhythmic behaviours that result from the coupling of environmental and endogenous factors (Herrnkind, 1983; Aréchiga and Rodríguez-Sosa, 1997; Meyer-Rochow, 1998). For example, spiny lobsters usually emerge from their dens to forage shortly after dusk. This rhythmic locomotor activity is coupled with photoperiod and may vary seasonally (Fielder, 1965; Paterson, 1969; Kanciruk and Herrnkind, 1973; Morgan, 1978; Herrnkind, 1980; Lipcius and Herrnkind, 1982). However, some palinurids are not exclusively nocturnal and may display activity in open areas during the daytime (e.g., *Jasus lalandii*: Paterson, 1969; *J. edwardsii*: McKoy and Leachman, 1982; *P. marginatus*: MacDonald *et al.*, 1984; *P. argus*: review in Herrnkind *et al.*, 2001). This does not seem to be the case for *P. guttatus*. Individuals of *P. guttatus* have not been observed outside their shelters

during the day, neither in the field (Sharp *et al.*, 1997) nor in the laboratory (Lozano-Álvarez and Briones-Fourzán, 2001).

Given the highly residential nature and small home range of *P. guttatus* as well as the great amount of time that this species remains in dens, we were interested in describing several issues of its within-shelter behaviour. For example, in a communal den, do individuals remain mostly quiescent or do they exhibit some level of activity? Is there any evidence of rhythmic behaviour in denned *P. guttatus*? We also investigated the tendency of individuals to aggregate or to remain solitary as well as the within-shelter behavioural patterns and sociality of groups of *P. guttatus*.

The coral reef is a highly intricate habitat, making it difficult to observe groups of individuals in their natural dens. For this reason, we conducted observations on groups of lobsters in simulated caves in tanks. However, because these observations were unreplicated due to restricted collections of lobsters, we present our results as descriptive and indicative of trends rather than conclusive.

METHODS

General Methods

Individuals of *P. guttatus* were captured with traps deployed by SCUBA divers beneath ledges and undercuts in the coral reef at Puerto Morelos, México (20° 51' N, 86° 55' W), at a depth of 5–6 m. In the morning, the divers carefully extracted the lobsters from the traps and introduced them into a seawater container in a boat to transport them to our laboratory. There, lobsters were transferred into a holding tank within 1 h of being captured. Lobsters remained in the holding tank for 2 weeks on average, a period that allows spiny lobsters to adapt to captivity conditions (Dall, 1975). During the acclimatisation period, shelter was provided in the form of low dens (flat-roofed “casitas”, Lozano-Álvarez and Briones-Fourzán, 2001), and lobsters were fed daily at 17:00 h with clams and mussels. Food remains were extracted every morning.

Outdoor circular tanks 3 m in diameter and 0.9 m in depth were used throughout the study. A continuous seawater flow maintained ambient water temperature in the tanks, which were under a translucent shade and hence subject to ambient light. In series 1 and 2 (see below), the tank held a triple-chamber artificial cave (Fig. 1), simulating an abundant shelter situation. The cave was constructed with three inverted plastic trash bins, 50 × 35 × 70 cm each, attached to two parallel PVC pipes, 7.5 cm in diameter, filled with lead to increase stability. This allowed lobsters to move freely between chambers without being exposed to the open. To provide gripping surface for the lobsters, the interior of each bin was lined with wire mesh slightly separated from the walls of the bin (Fig. 1). The top of each bin had two holes. A black-and-white surveillance video camera (Sony CC-2), encased in an underwater Plexiglas housing (Ikelite 5210), was attached to one of the holes in each bin. The other hole held an underwater lamp that produced a dim red light, to which lobsters in general show little response (Phillips *et al.*, 1980), for night recordings. The three cameras were connected to a multiplexor (Panasonic WJ-FS216) that allowed the simultaneous recording of the inside of the three cave chambers on a time-lapse

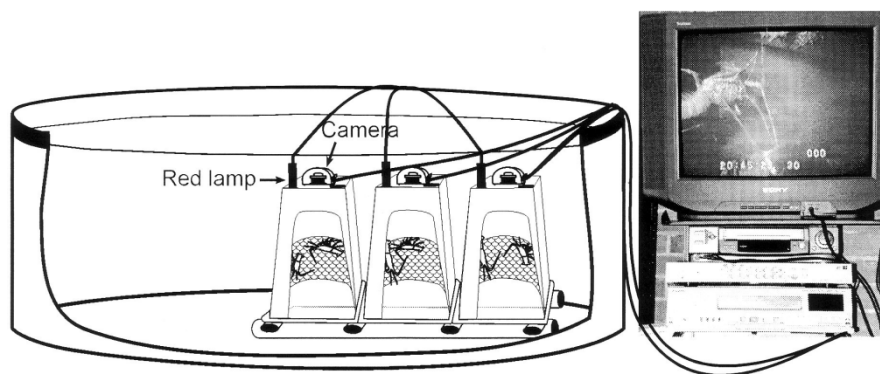


FIGURE 1 Schematic representation (not to scale) of the experimental tanks, the recording equipment, and the triple-chamber cave used in series 1 and 2. Chambers of cave are “cut” longitudinally to show interior. In series 3, the cave had only one chamber.

VCR (Sony SVT-L400). In series 3, the cave consisted of a single chamber, simulating a limited shelter situation.

To establish a behavioural baseline, we constructed an ethogram describing the within-shelter behavioural units of individual *P. guttatus* through the complete and detailed examination of 40 h of tape recordings of groups of four to six individuals within single-chamber caves. The tapes were recorded in December 1999, April, July, and September 2000, to encompass possible seasonal changes in behaviour. Thirty behavioural units were observed (Table I). Most matched those described by Atema and Cobb (1980) and Cobb (1981) for *P. cygnus* and by Roth (1972) and Krekorian *et al.* (1974) for *P. interruptus*, but a few had not been previously described (Table I). The within-shelter behavioural units were then grouped in four behaviour categories: (1) resting (individuals at rest, not involved in any other type of behaviour); (2) grooming (cleaning of body parts with appendages); (3) locomotory (movements of individuals not related to agonistic encounters); and (4) agonistic (agonistic interactions involving two or more individuals) (Table I).

Prior to each observational series, lobsters were measured (CL from between the supraorbital horns to the posterior margin of the cephalothorax) with calipers (± 0.1 mm) and their sex was determined. Each lobster was distinctively marked with contrasting dark and clear adhesive tapes glued to its carapace. Only lobsters in intermoult were used, because moult stage is known to affect the behaviour and diel rhythms of spiny lobsters (Morgan, 1978; Lipcius and Herrnkind, 1982). The composition of the groups of lobsters used in each series was imposed by the restricted collection of individuals and the number of lobsters in intermoult at the end of the acclimatisation periods. Feeding of lobsters was suspended two days before the beginning of experiments and lobsters were not fed during the experimental periods.

Series 1: Within-shelter Activity Rhythm

A group of six individuals, two females and four males (mean CL $\pm$ SD: 60.7 ± 6.6 mm), was used to study the within-shelter activity rhythm in a tank holding a triple-chamber cave. Recordings were conducted for 5 min every hour (chosen at random from the

TABLE 1 Behavioural units of *Panulirus guttatus*, observed in individuals within shelters. Categories “resting”, “grooming”, and “locomotory” apply to one individual (not involving another lobster). Category “agonistic” involves two or more individuals. Only the definition of previously undescribed units (with an asterisk) is provided

Behaviour category	Behavioural unit	Description
Resting	Resting	As in Cobb (1981).
	Leg swing	As in Cobb (1981).
	Splay	As in Cobb (1981).
Grooming	*Antennule cleaning	Abdomen tucked. Pereiopods directed forward; antennule lowered and grasped by third maxillipeds. Antennule slowly pulled upwards between maxillipeds.
	*Antenna cleaning	Body posture intermediate; abdomen extended or arched; pereiopods close to body. Dactyls of third or fourth pereiopods slide along the antenna, which is parallel to the side of the body.
	*Abdomen cleaning	Body posture intermediate; abdomen curled or arched; uropods extended; tailfan vertical. Dactyl of fifth pereiopods “scrub” abdominal tergites and sternites and uropods.
	*Carapace cleaning	Body posture intermediate; abdomen curled; dactyl of third or fourth pereiopods reach above to “scratch” dorsal surface of carapace.
	*Eggmass cleaning	Ovigerous females softly agitate pleopods. Subchelae of fifth pereiopods “pick” on the eggs, removing debris and (presumably) empty egg capsules.
	*Pereopod cleaning	All pereiopods in low posture, directed forward; dactyls of each pair “scrub” dactyls of paired pereopod.
Locomotory	Walk	As in Atema and Cobb (1980) and Cobb (1981), and as “roaming” in Krekorian <i>et al.</i> (1974).
	Explore	As in Atema and Cobb (1980) and Cobb (1981).
	*Detach	A lobster clinging to the wall or roof of cave suddenly contracts and extends all pereopods and lets go, sinking to a lower part or all the way to the floor of the cave.
Agonistic	Antenna contact	As in Atema and Cobb (1980) and Cobb (1981).
	Antennule contact	As in Atema and Cobb (1980) and Cobb (1981).
	Antenna point	As in Atema and Cobb (1980) and Cobb (1981).
	Antennule point	As in Atema and Cobb (1980) and Cobb (1981).
	Approach	As in Krekorian <i>et al.</i> (1974), Atema and Cobb (1980), and Cobb (1981).
	Avoid	As in Atema and Cobb (1980) and Cobb (1981).
	Clasp	As in Roth (1972), Atema and Cobb (1980), and Cobb (1981).
	Chase	As in Roth (1972) and Krekorian <i>et al.</i> (1974).
	Crouch	As in Atema and Cobb (1980) and Cobb (1981).
	Fight	As in Atema and Cobb (1980) and Cobb (1981).
	Flee	As in Cobb (1981).
	Frontal approach	As in Cobb (1981).
	Grasp	As in Atema and Cobb (1980) and Cobb (1981).
	Probe	As in Cobb (1981).
	Push	As in Cobb (1981).
	Rush	As in Atema and Cobb (1980) and Cobb (1981).
	Tail flip	As in Atema and Cobb (1980) and Cobb (1981).
	Walk away	As in Krekorian <i>et al.</i> (1974), Atema and Cobb (1980), and Cobb (1981).

first half of the hour) during five continuous 24-h cycles (01:00 h of April 29 to 00:00 h of May 4, 2000), and observed to quantify the number of movements of individual lobsters within the cave (Martin and Bateson, 1993). “Movement” was defined as a lobster moving a distance longer than or equal to its body length. The total number of movements during each 5-min recording was divided by the number of lobsters that were within the cave to obtain a “mean index of within-shelter activity”. Rhythmicity in

within-shelter activity was assessed with a Fourier analysis of hourly indices, followed by harmonic analysis. A Friedman test (Zar, 1999) was applied to the number of movements of each lobster over the five 24-h cycles to determine whether the hourly index of activity varied significantly over time.

Series 2: Den Occupancy Patterns and Within-shelter Aggressive Interactions

The experimental tank in series 2 held a triple-chamber cave similar to the one used in series 1. The group of lobsters consisted of five individuals: a large male (71.0 mm CL); a nonovigerous female (52.3 mm CL); an ovigerous female (56.8 mm CL); an intermediate male (62.7 mm CL), and a small male (58.3 mm CL).

Five different recordings (each 5-min long) for every hour in a 24-h cycle were conducted between May 9 and May 19, 2000. Aggressive interactions between individuals within the shelter were scored in all recordings. One instantaneous sampling (Martin and Bateson, 1993) was then performed at random within each 5-min recording to score the number of individuals within and outside the triple-chamber cave throughout the 24-h cycle. We then scored the number of individuals in each of the three chambers during the five recordings in each of the 12 h of daylight (07:00 to 18:00 h, when all individuals were sheltered) and assessed their tendency to aggregate (i.e., three or more individuals in close proximity within the same chamber) by testing the results with a goodness-of-fit test against all possible combinations of numbers of individuals per chamber (Zar, 1999). We also assessed the preference of individuals to stay on the floor or to cling to the walls with contingency tables (Zar, 1999).

Series 3: Within-shelter Behaviour of a Group of Lobsters

Based on the ethogram (see Table I) and the results of the within-shelter activity rhythm (series 1), we analysed a tape that recorded the behaviour of a group of four individuals, one male and three females (mean $\pm$ SD: 58.3 $\pm$ 4.2 mm CL), in a single-chamber cave, from 19:00 to 06:00 h of the following day (14–15 April, 2000), i.e., when the index of within-shelter activity was highest. The single-chamber cave simulated a more limited shelter situation, because in reef substrates large dens are scarcer than small dens owing to the fractal surface of coral reefs (Caddy, 1986). Instantaneous samplings of the behavioural units described in Table I were made for each of the four individuals every 5 min, giving a total of 48 units sampled per hour (4 individuals $\times$ 12 sampling points per hour). We then plotted a composite graph depicting the hourly percent of occurrence of each behavioural category (see Table I) performed by the group of four lobsters.

RESULTS

Series 1: Within-shelter Activity Rhythm

Lobsters displayed within-shelter activity throughout the day, but the index of within-shelter activity was higher during the night than during the day. In general, within-shelter activity peaked prior to dawn and was lowest around noon (Fig. 2). The periodogram resulting from the Fourier analysis revealed a strong 24 h rhythm

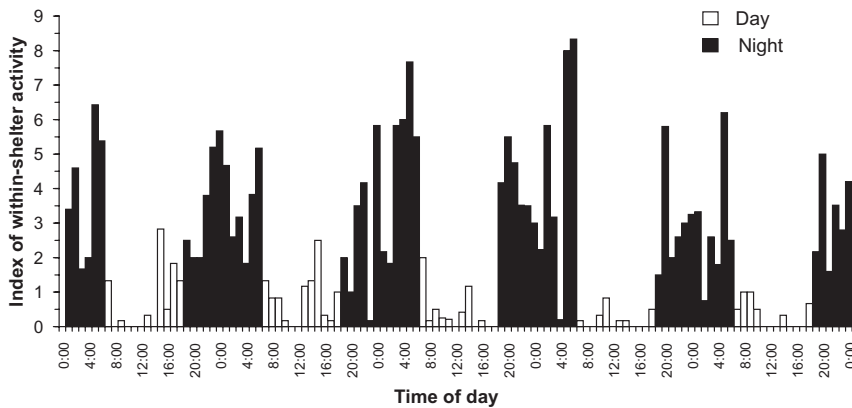


FIGURE 2 Index of activity (mean number of within-shelter movements) of *Panulirus guttatus* in a triple-chamber cave, recorded throughout 5 min per hour from 01:00 h of April 29 to 00:00 h of May 4, 2000 ($N = 6$ lobsters).

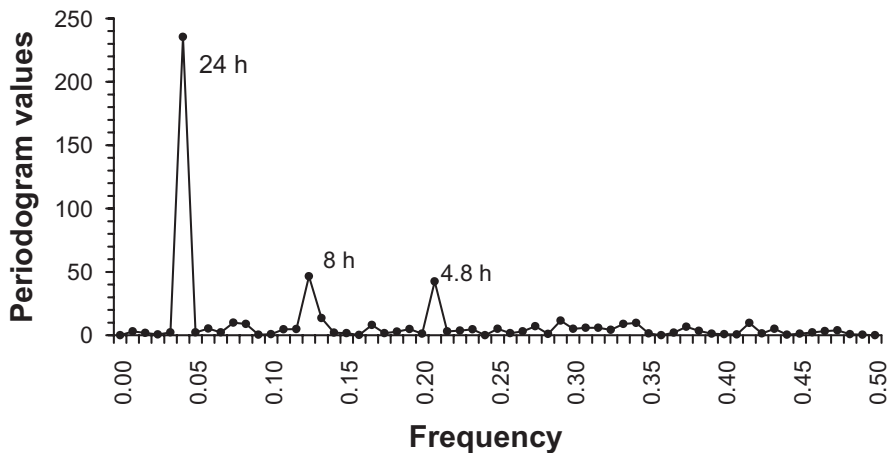


FIGURE 3 Periodogram resulting from Fourier analysis of hourly index of within-shelter activity of a group of six lobsters in a triple-chamber cave during five 24-h cycles. Peaks in the periodogram are labelled with period of rhythm.

in within-shelter activity (Fig. 3). Smaller secondary rhythms also occurred, but harmonics showed that the 24 h period in within-shelter activity explained 44% of the variance.

The mean index of within-shelter activity (Fig. 4) varied significantly throughout the 24-h cycle (Friedman test: $\chi^2_r = 91.969$, $df = 23$, $P < 0.001$). Kendall coefficient of concordance (0.680) indicated a significant agreement ($P < 0.001$) of hourly activity rankings among individuals. An *a posteriori* Tukey test on ranked data ($\alpha = 0.05$) separated the activity data in two periods, one encompassing from 19:00 to 06:00 h, and the other one from 07:00 to 18:00 h. Therefore, the period of highest within-shelter activity was from 19:00 to 06:00 h, with a major peak around dawn and a lesser peak soon after dusk.

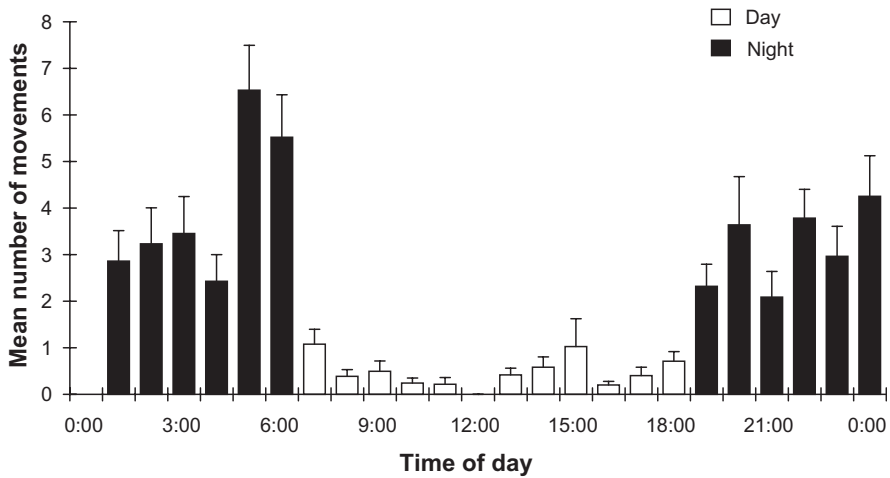


FIGURE 4 Hourly mean index of within-shelter activity of a group of six individuals of *Panulirus guttatus* in a triple-chamber cave (vertical lines represent 1 SE of the mean). Index of activity (mean number of within-shelter movements) was quantified throughout 5 min per hour, from 01:00 h of April 29 to 00:00 h of May 4, 2000.

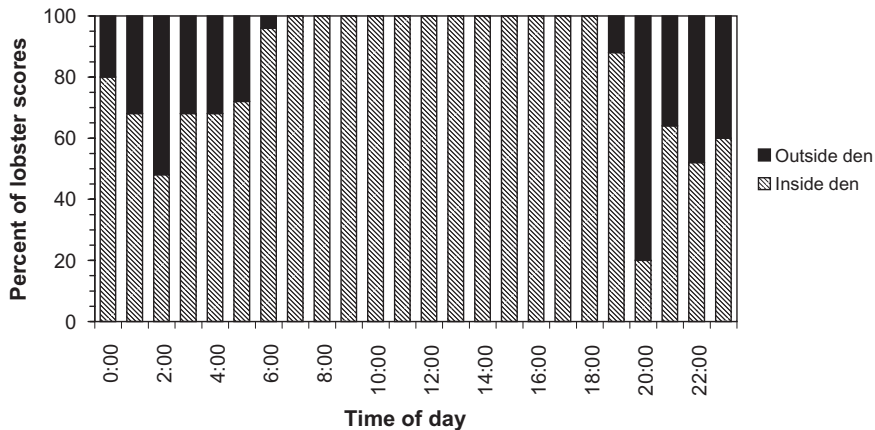


FIGURE 5 Hourly den occupancy of five individuals of *Panulirus guttatus* in a triple-chamber cave. The occurrence of each lobster inside or outside the den was scored once every hour during five 24 h periods ($N=25$ scores per hour of day).

Series 2: Den Occupancy Patterns and Within-shelter Aggressive Interactions

The lowest percentage of lobsters within the den occurred at 20:00 h, but at no time at night were all lobsters outside the cave and many were inside the cave for a variable portion of the night (Fig. 5). In contrast, all lobsters were always inside the cave between 07:00 and 18:00 h.

There were five possible combinations of the number of lobsters sheltered in any one of the three chambers (Table II). Assuming an equal probability of occurrence

TABLE II Distribution of five individuals of *Panulirus guttatus* in any one of the three chambers of the cave, in scores taken once every hour of daylight (07:00–18:00 h) during five different periods ($N=60$ scores)

Number of lobsters per chamber	Number of occurrences
5–0–0	7
4–1–0	26
3–2–0	2
3–1–1	17
2–2–1	8
Total scores	60

to each of the five possible combinations, the goodness-of-fit test showed a significant departure from this probability of the 60 scores obtained ($\chi^2 = 30.167$, $df=4$, $P < 0.001$) (Table II). In contrast, scores including ≥ 3 lobsters in a chamber fitted the probability ($\chi^2 = 1.666$; $df=1$, $P > 0.25$), suggesting that individuals of *P. guttatus* tended to aggregate in any one chamber during the day. However, on 71% of the 60 scores the number of lobsters per chamber was 4–1–0 (26 scores) and 3–1–1 (17 scores) (Table II), indicating that the aggregation of the five individuals was unusual and that some individuals were solitary (alone in a chamber) a great number of occasions. The largest male was solitary on 26 (43%) of the scores; the intermediate male on 20 (33.3%); the nonovigerous female on 19 (31%); the smallest male on 2 (3%), and the ovigerous female on 1 (1.7%).

The occurrence of solitary individuals was often, but not always, an outcome of aggressive interactions. Of the 44 aggressions recorded, two occurred during the day and 42 throughout the night hours, with 10 (24%) occurring prior to dawn. The largest male was the actor in 42 (95%) aggressions, which were directed at conspecifics entering or already within the chamber he occupied. Many aggressive interactions resulted in the eviction of the recipient from that particular chamber. As a consequence, both the actor and the recipient often ended up alone in different chambers. The recipient would eventually move to the chamber containing the rest of the lobsters, unless the evictions occurred at dawn, in which case the recipient (and the actor) would remain solitary during the remaining day hours, when movements between chambers were negligible. Whether aggregated or solitary, individuals showed a strong preference to cling to the walls of caves rather than to staying on the floor (300 scores: 283 on walls, 17 on floor; $\chi^2 = 235.853$, $df=1$, $P < 0.0001$).

Aggressions by the largest male were not equally directed at its conspecific (19 at the smallest male, 10 at the intermediate male, 8 at the non-ovigerous female, and 5 at the ovigerous female; $\chi^2 = 10.381$, $df=3$; $P < 0.025$). Clearly, the smallest male was the recipient of the most aggressions. When this male was excluded from the test the number of aggressions directed by the largest male at the other three individuals was similar ($\chi^2 = 1.668$, $df=2$; $P > 0.25$).

Series 3: Within-Shelter Behaviour of a Group of Lobsters

The within-shelter behaviour of the group of four lobsters in the single-chamber cave is depicted in Fig. 6. Soon after dusk (19:00 h), some individuals left the den while the remaining individuals rested or groomed – especially their antennules—apparently in preparation for a foraging excursion. From 20:00 to 22:00 h most individuals

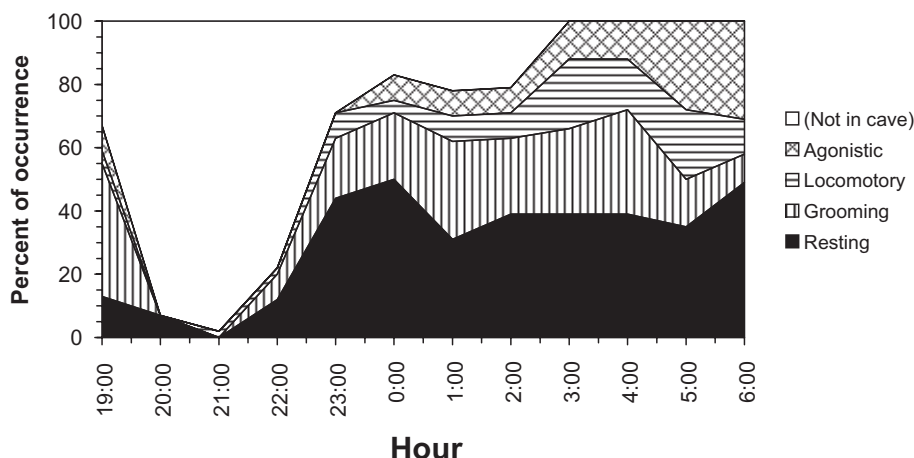


FIGURE 6 Hourly percent of behaviour categories displayed by a group of four *Panulirus guttatus* within a single-chamber cave, from 19:00 h of 14 April to 06:00 h of 15 April, 2000. Behaviour of each lobster was scored 12 times every hour ($N=48$ behaviours per hour). Lobsters outside den were out of the scope of the camera and were thus recorded as “not in cave”. See Table I for description of categories.

were outside the cave and hence out of the scope of the camera. From 23:00 to about 02:00 h, some individuals entered and emerged from the cave several times; in this period, about 40% of the behaviour scores within the cave involved resting and 25% grooming of different body parts, whereas the incidence of locomotory activity or agonistic displays was low. However, by 03:00 h, long before sunrise, all the lobsters were back in the cave and none emerged from it again. Thereafter, grooming decreased and agonistic interactions increased to a peak of 32% at 06:00 h. Agonistic interactions resulted in the constant rearrangement of individuals within the cave.

DISCUSSION

Our results show that *P. guttatus* is exclusively nocturnal, as previously suggested by Sharp *et al.* (1997) and Lozano-Álvarez and Briones-Fourzán (2001). In our different series, not all individuals emerged from the dens at the same time, and some went in and out of the dens several times during the night, but all the lobsters were generally back in the cave before sunrise and remained sheltered until the following night. Differences in the amount of outside-shelter activity during the day among spiny lobster species may be related to different defensive strategies. For example, the communal defensive displays shown by *P. argus* in open areas during the day, which include the formation of pods, rosettes or queues (review in Herrnkind *et al.*, 2001), have never been observed in *P. guttatus*. The defensive strategy of *P. guttatus* rather involves hiding deeply in dens during the day, preferring shelters where predators find it difficult to enter, and deterring those that do enter with their long and spiny antennae (Lozano-Álvarez and Briones-Fourzán, 2001).

As is common for many nocturnal crustaceans (Aréchiga and Rodríguez-Sosa, 1997; Meyer-Rochow, 1998), slow changes in light intensity that occur at dusk probably govern the daily emergence of individuals of *P. guttatus* from their dens, but groups

of *P. guttatus* also showed a diel activity rhythm within the shelter. The major peak in within-shelter activity consistently occurred around dawn, which may be similarly coupled with changes in light intensity, but was also related to an increase in agonistic interactions among sheltered individuals. In our series 2, this resulted in some individuals ending up alone in one of the three chambers. In our series 3, where only one chamber was available, the increase in agonistic interactions prior to dawn resulted in the constant rearrangement of individuals within the cave, which in turn appears related to the preference of *P. guttatus* to cling to the walls of caves rather than to remaining on the floor.

The preference of *P. guttatus* to cling to the walls of dens, irrespective of the presence in the same caves of *P. argus* or other crevice-dwelling reef organisms, had been previously documented in the field (Sharp *et al.*, 1997) and in the laboratory (Lozano-Álvarez and Briones-Fourzán, 2001). In observations conducted in single-chamber caves deployed in tanks that held a mixed group of *P. guttatus*, *P. argus*, and one diurnal fish predator, Lozano-Álvarez and Briones-Fourzán (2001) found that individuals of *P. guttatus* would climb higher than those of *P. argus* and that the largest individuals of *P. guttatus* would defend the vault of the caves from their smaller conspecifics. Therefore, the highest part of a single-chamber den is presumably the safest place for *P. guttatus*, and individuals with the highest aggressive rank will tend to actively defend this place from other conspecifics, which would be displaced to other, presumably less favorable spots in the cave.

In our series 2, where a triple-chamber cave was used, the largest male would often attack other individuals inside a chamber. Berrill (1975) described a similar behaviour between juvenile *P. argus*, which resulted in the eviction from the den of any of the two individuals involved. By contrast, the evicted individual in our study was always the smaller of the two contenders. High aggressive rank, often coupled with size, confers advantage in gaining shelter and/or food upon animals, at least in laboratory situations (Fielder, 1965; Roth, 1972; Wilson, 1975; Cobb, 1981), and in our study the most aggressive individual was the largest male, similar to the findings of Berry (1970) in *P. homarus*, Roth (1972) in *P. interruptus*, and McKoy (1979) in *Jaes edwardsii*. In Florida reefs, males of *P. guttatus* were observed seemingly guarding den entrances to protect "harems" of females from other males (Sharp *et al.*, 1997). If so, then the largest male in our triple-chamber cave simulation may have been defending a potential den in which to later mate and guard females. In fact, interactions recorded as aggressions at the two females may have been confounded with sexual approaches, which are very similar to agonistic encounters (Lipcius *et al.*, 1983), but no attempts to mate were discernible in our relatively short observation periods.

The occurrence of solitary and aggregated individuals in dens is common in other spiny lobsters that have been experimentally shown to be gregarious, such as *P. argus* (Lipcius and Herrnkind, 1985), *P. cygnus* (Cobb, 1981), and *P. interruptus* (Zimmer-Faust and Spanier, 1987). In our series 2, when one or two lobsters were alone in a chamber the remaining individuals tended to aggregate in one of the other two chambers. Thus, the trends in our results suggest that *P. guttatus* is also gregarious, but that the distribution of individuals among dens may be influenced by the availability of dens and by temporary, size-related social hierarchies, a situation that is probably general for spiny lobsters. Fully replicated experiments are needed to further test this hypothesis. Also, it remains to be determined whether gregariousness in *P. guttatus* is mediated by intraspecific chemical attraction, as in *P. interruptus*

(Zimmer-Faust *et al.*, 1985), *P. argus* (Ratchford and Eggleston, 1998), and *J. edwardsii* (Butler *et al.*, 1999).

The social behaviour of spiny lobsters has been the subject of several studies (e.g., Roth, 1972; Krekorian *et al.*, 1974; Atema and Cobb, 1980; Herrnkind *et al.*, 2001), but other types of behaviour, especially when lobsters are sheltered, have not been equally addressed. To our knowledge, ours is the first study categorising the behaviour of a group of lobsters within a communal den. Obviously, spiny lobsters are not simply quiescent when denned. Aside from den exploring activities and social interactions, bouts of antennular, body, and antennal grooming – occurring in that order in our series 3 – appear to occupy a great amount of time. Antennular grooming is important because these appendages contain the chemosensory setae that mediate perception of food and pheromones at a distance (Bauer, 1989; Daniel *et al.*, 2001), but other types of sensory sensilla occur over the cuticle of palinurids (Phillips *et al.*, 1980; Derby *et al.*, 2001) and need to be constantly cleaned as well.

All the individuals of *P. guttatus* in our study were sexually mature adults. Because *P. guttatus* dwells solely in the coral reef habitat during its entire benthic life, it would appear unlikely for this species to exhibit ontogenic changes in activity and behaviour associated with habitat shifts, such as those described for other, more mobile palinurids (review in Lipcius and Eggleston, 2000). However, juveniles of *P. guttatus* are seldom observed roaming at night in the coral reef (Sharp *et al.*, 1997); therefore, juveniles and adults may differ in their foraging activities as well as in their patterns of behaviour and shelter utilisation. These issues should be further investigated to increase the life-history knowledge of this species.

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